

• **What is Serial Processing in Computer Architecture**

In serial processing, the processor completes one task at a time. After completing that, it executes the other tasks in a sequential manner. An operating system executes many programs and each of them has multiple tasks. The processor has to complete all these tasks, but it completes one task at a time. The other tasks wait in the queue until the processor completes the current task. In other words, all tasks are processed sequentially. Hence, this type of processing is called serial processing or sequential processing. Machines such as Pentium 3 and Pentium 4 perform serial processing.

- ❖ We can understand the functionality of serial processing using the following analogy. Assume a supermarket with multiple queues and only one cashier. The cashier finishes billing the products of one customer and then moves on to another customer. He performs billing one after the other.

• **What is Parallel Computing?**

Parallel Computing is the process of running multiple computational tasks simultaneously by delegating different parts of the computation to different processors that execute at the same time. A parallel program uses several processor cores to perform a computation more quickly. It physically runs parts of tasks or multiple tasks at the same time using multiple processors. It enables single sequential CPUs to do lot of things seemingly simultaneously. Parallelism is a realization of a concurrent program.

In a multi-core system, multiple programs can actually make progress at the same time without the aid of the operating system to provide time slicing. If you run, let's say two processes on a dual-core system and allocate one core per process, they will both execute at the same time. This is what you can call executing in parallel. Parallel computing essentially requires hardware with multiple processing units. It is the simultaneous execution of computations, possibly related but not necessarily.

- ❖ We can understand the functionality of parallel processing using the following example. In a supermarket, there are multiple queues, and there is a cashier for each queue. Each cashier bills the products of the customers in his own queue.

- **Difference Between Serial and Parallel Processing in Computer Architecture**

Definition

Serial processing is a type of processing in which one task is completed at a time and all the tasks are executed by the processor in a sequence. Parallel processing is a type of processing in which multiple tasks are completed at a time by different processors. Thus, this is the fundamental difference between Serial and Parallel Processing.

Number of processors

A major difference between serial and parallel processing is that there is a single processor in serial processing, but there are multiple processors in parallel processing.

Performance

Therefore, the performance of parallel processing is higher than in serial processing.

Work Load

In serial processing, the workload of the processor is higher. However, in parallel processing, the workload per processor is lower. Thus, this is an important difference between serial and parallel processing.

Data transferring

Moreover, in serial processing, data transfers are in bit by bit format. However, in parallel processing, data transfers are in byte form (8 bits).

Required time

Time taken is also a difference between serial and parallel processing. That is; serial processing requires more time than parallel processing to complete a task.

Cost

Furthermore, parallel processing is more costly than serial processing as it uses multiple processors.

Conclusion

There are two types of processing as serial and parallel processing in a computer system. The main difference between serial and parallel processing in computer architecture is that serial processing performs a single task at a time while the parallel processing performs multiple tasks at a time. In brief, the performance of parallel processing is higher than the serial processing.

• **What is Concurrent Computing?**

Concurrent Computing is the concurrent (simultaneous) execution of multiple computational tasks in overlapping time periods instead of sequentially. It is the ability of a system to perform several calculations simultaneously or within overlapping time frames. Concurrency is essentially applicable when you are talking about more than one task at the same time. These tasks may be implemented as separate programs, or as a set of processes or threads created by a single program. The tasks may be executed on a single processor, multiple processors, or distributed across a network.

Concurrent computing is related to parallel computing, but focuses more on the interactions between tasks. Concurrency refers to the execution of multiple tasks at the same time but does not necessarily mean simultaneously. Whether the tasks accomplished at the same time or not is an implementation detail. The task can execute on a single processor through interrupted execution or on multiple physical processors. A common example of concurrency is a program to calculate the sum of a large list of numbers.

Concurrent Computing	Parallel Computing
Two or more computational tasks run and execute in overlapping time periods instead of sequentially.	Multiple computational tasks are executed simultaneously to solve a computational problem.
Tasks may be executed on a single processor, multiple processors, or distributed across a network.	Physically runs parts of tasks or multiple tasks at the same time using multiple processors.
Concurrency is the composition of executing processes independently.	Parallelism is the simultaneous execution of multiple computational tasks.
Concurrent programs are often non-deterministic because the program may give different results, depending on some aspect of execution.	Parallel programs use deterministic models in which each program can give only one result. 

❖ Summary

In a nutshell, concurrent computing means a program or task can support multiple computations at the same time, but not necessarily simultaneously. It is the process of performing computations independently. Parallel computing, on the other hand, refers to the simultaneous execution of two or more computations on different processors. You can say that all parallel computing is concurrent, but not the other way around. Parallel computing is not possible with single CPU; instead, it requires multi-core setup.